

Exam 1 Review Problems

STAT 251 SECTION 01

2026-09-16

These problems cover everything on Exam 1: Weeks 1 through 3, up to and including cumulative distributions. They are **more computational than the exam** by design. The exam asks you to interpret values, identify equations and read plots; these ask you to work the arithmetic behind them.

Work them by hand. Every dataset is small enough to do on paper, and a calculator is fine.

Problem 1 — The anatomy of data

A researcher recorded the following information for each of six trees in a city park.

Tree ID	Species	Health rating	Branches removed	Trunk width (cm)
T-01	Maple	Good	3	31.4
T-02	Oak	Fair	0	58.2
T-03	Maple	Excellent	5	22.7
T-04	Pine	Good	1	44.0
T-05	Oak	Poor	8	61.9
T-06	Maple	Good	2	27.5

(a) How many observations are in this data table, and how many variables?

(b) What is the value of n for this sample?

(c) Classify each of the four measured variables as nominal, ordinal, discrete or continuous.

Variable	Your answer
Species	
Health rating	
Branches removed	
Trunk width (cm)	

(d) These six trees are a sample. Describe the population the researcher is presumably interested in, and say whether N is known.

Problem 2 — Statistics, parameters, and inference

A city has 8,500 registered bicycles. A researcher takes a random sample of 200 of them and finds that 46 have no working lights. She also measures the age of each sampled bicycle and finds a mean age of 6.2 years.

(a) Give the values of N and n .

(b) Compute the sample proportion of bicycles with no working lights. Which symbol denotes it?

(c) For each quantity, say whether it is a **statistic** or a **parameter**, and give the symbol that denotes it.

Quantity	Statistic or parameter	Symbol
The mean age of the 200 sampled bicycles		
The mean age of all 8,500 registered bicycles		
The proportion of the 200 sampled with no lights		
The proportion of all 8,500 with no lights		

(d) The researcher reports the mean age of the 200 bicycles she measured, then uses it to estimate the mean age of all 8,500. Which of those two acts is descriptive statistics and which is inferential? Explain what makes the difference.

Problem 3 — Reading the formulas

On the exam the formulas are provided **unlabeled** — the equations only, with no names. This problem is practice at telling them apart.

(a) Each equation below computes one of the quantities in the list. Write the name of the quantity next to its equation.

sample mean · population mean · sample proportion · sample variance · sample standard deviation · interquartile range

Equation	What it computes	Symbol
$\frac{1}{n} \sum_{i=1}^n x_i$		
$\sqrt{\frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2}$		
$Q3 - Q1$		
$\frac{1}{N} \sum_{i=1}^N x_i$		
$\frac{\text{number of observations in the category}}{n}$		
$\frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$		

(b) (*notation*) Answer each in a few words.

Question	Your answer
What does n stand for?	
What does N stand for?	
What does the <i>hat</i> on $\hat{p}$ tell you?	
In $\sum_{i=1}^n x_i$, what does the $i = 1$ below the Σ mean?	

(c) Consider the following sample of five observations of a quantitative variable X :

$$X = \{2, 5, 6, 9, 13\}$$

Show how to compute the mean $\bar{x}$ and the standard deviation s .

(d) Now consider the following sample of ten observations of a qualitative variable Y , the way each of ten students travelled to campus:

$$Y = \{\text{Bus, Walk, Bike, Bus, Walk, Bus, Car, Bus, Walk, Bike}\}$$

Show how to compute the proportion of observations in the category **Bus**.

Problem 4 — The distribution of a qualitative variable

A campus coffee stand recorded the drink ordered by each of 20 customers one morning:

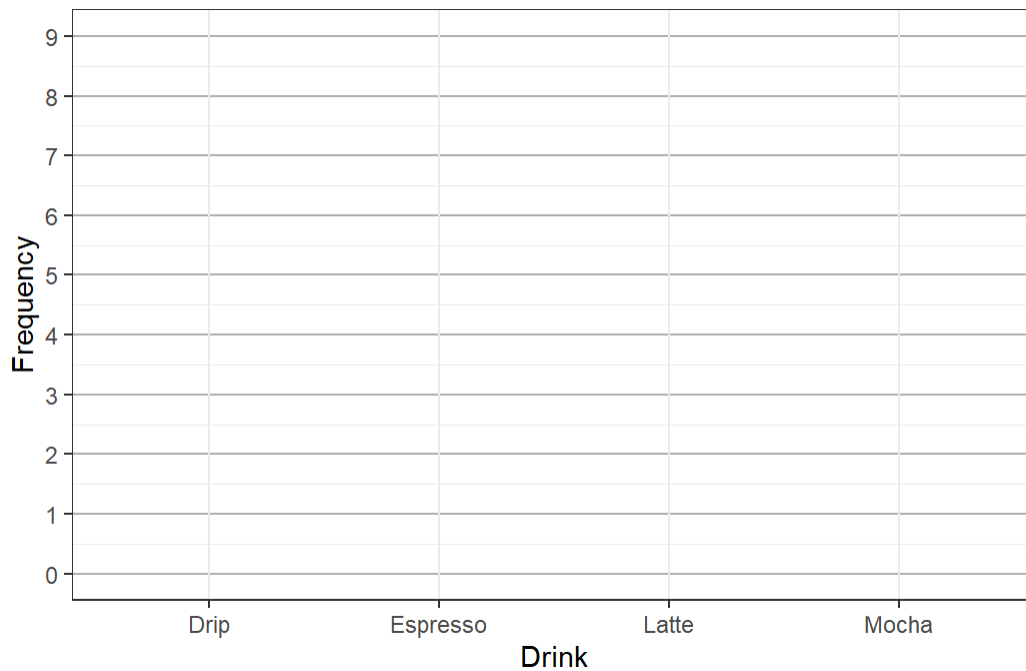
$$X = \{\text{Latte, Drip, Mocha, Latte, Drip, Espresso, Latte, Drip, Latte, Mocha, Drip, Latte, Espresso, Drip, Mocha, Latte, Drip, Latte, Mocha, Latte}\}$$

(a) Complete the frequency and relative frequency table.

Drink	Frequency $F(x)$	Relative frequency $RF(x)$
Drip		
Espresso		
Latte		
Mocha		

(b) What is the modal category?

(c) Draw a bar graph of the distribution on the axes below.



(d) Would it make sense to add a *cumulative relative frequency* column to the table in part (a)? Explain.

Problem 5 — Frequency tables and the mean from a table

Twenty-five households were asked how many vehicles they own. The responses are summarized below.

Vehicles x	Frequency $F(x)$	Relative frequency $RF(x)$	Cumulative relative frequency $CRF(x)$
0	3		
1	9		
2	8		
3	4		
4	1		

(a) Complete the relative frequency and cumulative relative frequency columns.

(b) Compute the mean number of vehicles using the **frequency** formula, $\bar{x} = \frac{1}{n} \sum_x x F(x)$.

(c) Now compute it again using the **relative frequency** formula, $\bar{x} = \sum_x x RF(x)$, and confirm you get the same answer.

(d) What proportion of these households own 2 or fewer vehicles? Which column did you read that from?

(e) What is the median number of vehicles owned? (You can read this off the cumulative relative frequency column.)

Problem 6 — Binning a continuous variable

The following are the times, in seconds, that 16 users took to complete a task on a website.

$$X = \{12.4, 14.1, 15.8, 16.2, 17.5, 18.0, 18.3, 19.1, 20.2, 20.6, \\ 21.7, 22.4, 23.9, 25.1, 26.8, 28.4\}$$

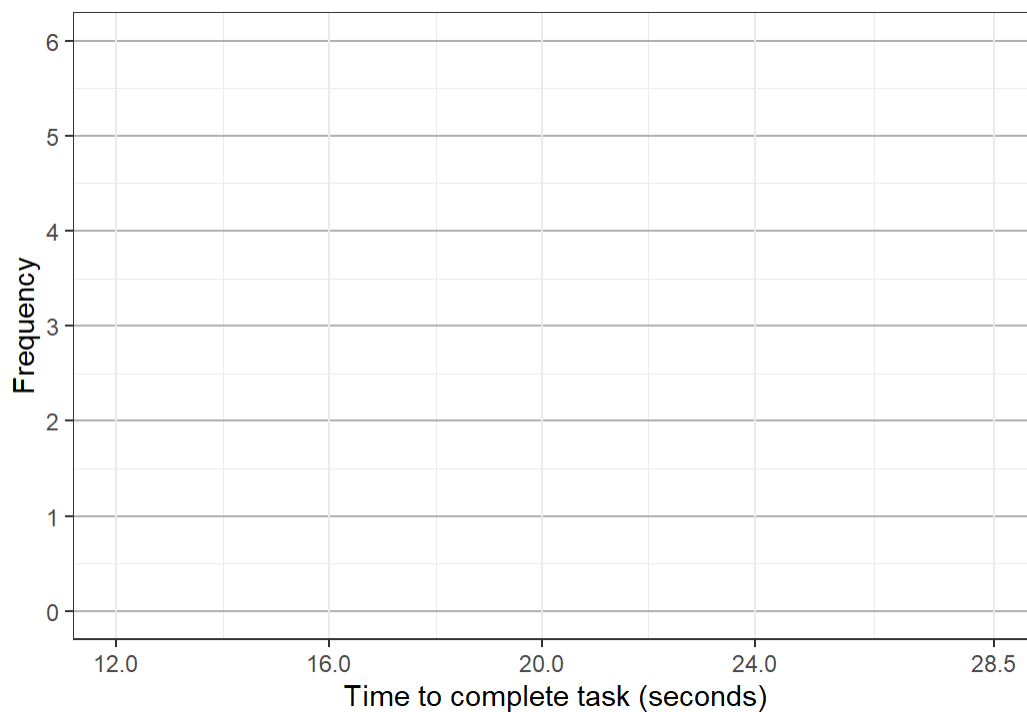
(a) Using the **square root rule**, how many bins k should you use for a histogram of these 16 observations?

(b) Compute the bin width w implied by that choice, using $w = \frac{\max x - \min x}{k}$.

(c) Complete the frequency table using the intervals given.

Time (seconds)	Frequency $F(x)$	Relative frequency $RF(x)$
$12 < x \leq 16$		
$16 < x \leq 20$		
$20 < x \leq 24$		
$24 < x \leq 28.5$		

(d) Draw the histogram on the axes below.



(e) Describe the shape of this distribution.

(f) Give one thing you can determine from the table in part (c), and one thing you **cannot**, that you could have determined from the raw data.

Problem 7 — Dot plots and stem-and-leaf plots

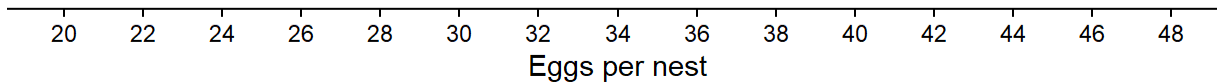
A biologist recorded the number of eggs in each of 15 birds' nests:

$$X = \{21, 24, 24, 27, 28, 31, 33, 33, 33, 35, 38, 41, 42, 44, 47\}$$

(a) Construct a **stem-and-leaf plot**, using the tens digit as the stem.

2 |
3 |
4 |

(b) Construct a **dot plot** on the number line below.



(c) What is the mode of this variable?

(d) Is the mode a measure of the center of this distribution? Explain.

Problem 8 — Center, and what an outlier does to it

Consider the following eight observations of a variable X , the number of minutes eight customers spent in a shop:

$$X = \{4, 6, 6, 9, 11, 12, 14, 16\}$$

(a) Compute the mean, the median and the mode.

Statistic	Your answer
Mean $\bar{x}$	
Median	
Mode	

(b) A ninth customer stays for **78 minutes**. Recompute the mean and the median with that observation included.

(c) Which statistic moved more, and why?

(d) Which of the two is **resistant**? Give a one-sentence definition of that word.

Problem 9 — Percentiles and quartiles

The following are the ages of the 11 members of a community board, sorted:

$$X = \{29, 34, 38, 41, 45, 49, 52, 56, 61, 67, 73\}$$

(a) Find the median, $Q2$.

(b) Find $Q1$ and $Q3$ using the **method of halves** we use in this class. Remember that when n is odd the median is **left out** of both halves.

(c) A twelfth member, aged 58, joins the board. Find $Q1$, $Q2$ and $Q3$ for the new set of 12 ages.

(d) What is the IQR , and what does it measure?

(e) The board member aged 67 is at roughly what percentile? Answer in a sentence that says what that means.

Problem 10 — Deviations, variance and standard deviation

A quality inspector measured the weight, in grams, of six packets from a filling machine:

$$X = \{48, 51, 52, 55, 57, 59\}$$

(a) Compute the sample mean $\bar{x}$.

(b) Complete the deviation table below, then add up the deviation column.

x_i	$x_i - \bar{x}$	$(x_i - \bar{x})^2$
48		
51		
52		
55		
57		
59		

(c) Compute the sample variance s^2 and the sample standard deviation s .

(d) State the units of $\bar{x}$, of s^2 and of s .

Quantity	Units
$\bar{x}$	
s^2	
s	

(e) Compute the range. Which is the better description of the spread of this data, the range or the standard deviation, and why?

Problem 11 — Five-number summary, boxplots and outliers

A clinic recorded the number of days between a referral and an appointment for 12 patients:

$$X = \{3, 5, 6, 8, 9, 11, 12, 14, 15, 17, 19, 58\}$$

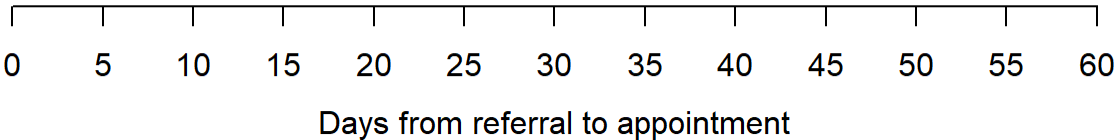
(a) Give the five-number summary. Use the method of halves for the quartiles.

Minimum	Q1	Median	Q3	Maximum

(b) Compute the IQR and both outlier boundaries.

(c) Which observations, if any, does the $1.5 \times IQR$ rule flag as outliers?

(d) Draw the boxplot on the axis below. Be careful about where the right whisker ends.



- (e) Describe the shape of this distribution from the boxplot, and say what that implies about the mean relative to the median.
- (f) The clinic manager wants to delete the 58-day wait because it is “clearly a mistake.” What would you tell her?
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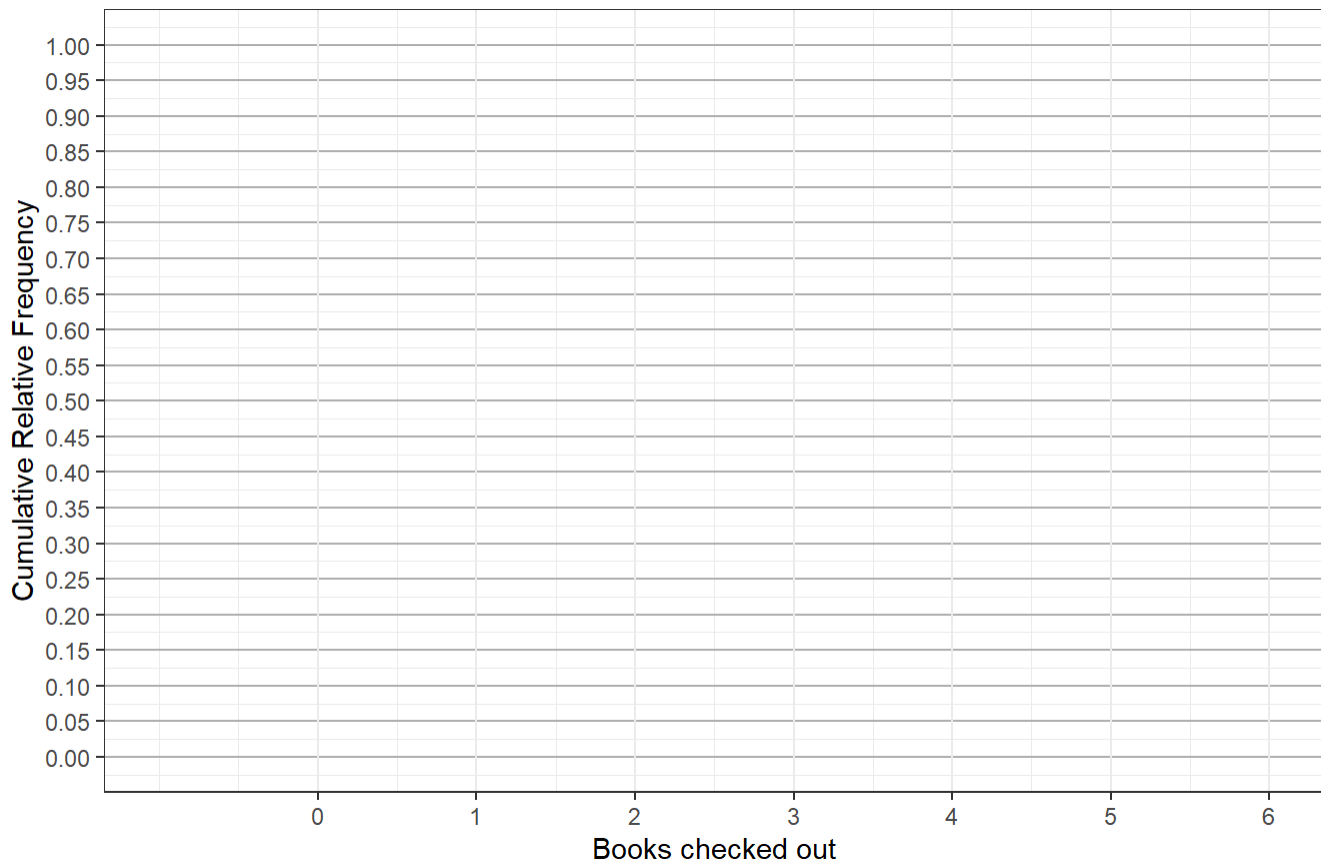
Problem 12 — Cumulative distributions

A librarian recorded the number of books checked out by each of 20 patrons in one hour.

(a) Complete the table.

Books x	Frequency $F(x)$	Relative frequency $RF(x)$	Cumulative relative frequency $CRF(x)$
0	2		
1	5		
2	6		
3	4		
4	2		
5	1		

(b) Plot the cumulative distribution on the axes below. Remember it is a **step function**: flat between observed values, jumping at each one.



(c) Use the plot to find Q_1 , the median, and Q_3 .

(d) What proportion of patrons checked out 2 or fewer books?

(e) Describe the shape of this distribution, and say whether the mean will be greater than, less than, or equal to the median.

Check your work

Worked solutions are posted under **Exam Study Guides → Exam 1 Review → Solutions**. Work each problem through before looking at them.

Every quartile on this exam uses the method of halves. Problem 9 works it both ways: with an odd n , where the median is dropped, and with an even n , where it is not.